

shallow deep shielding summary

Near-surface vs deep storage shielding

Near-surface = CONUS2 layers $z \geq 6$ (top 2 m, GWMM unconfined). Deep = $z < 6$.

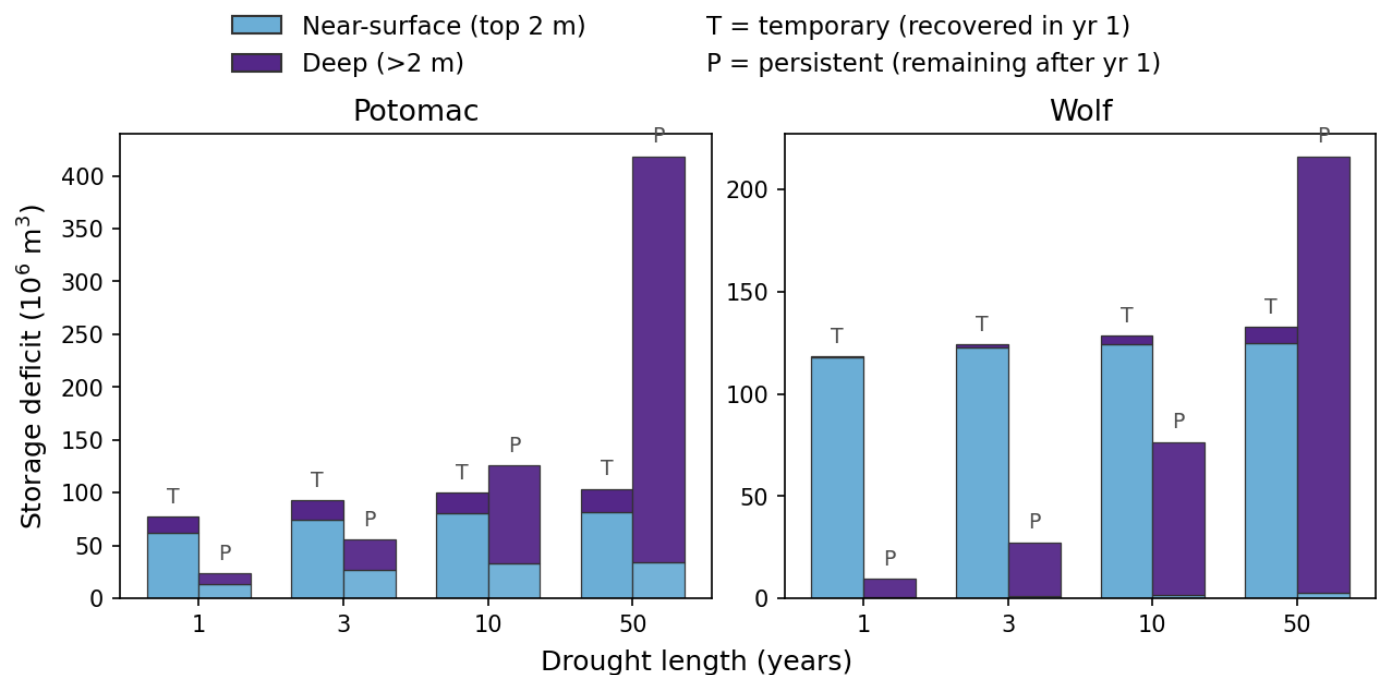
Claim tested: places whose drought losses stay near the surface regenerate those losses on precip pulses *during* drought, so the losses remain temporary and the column is shielded from persistent (deep) drawdown.

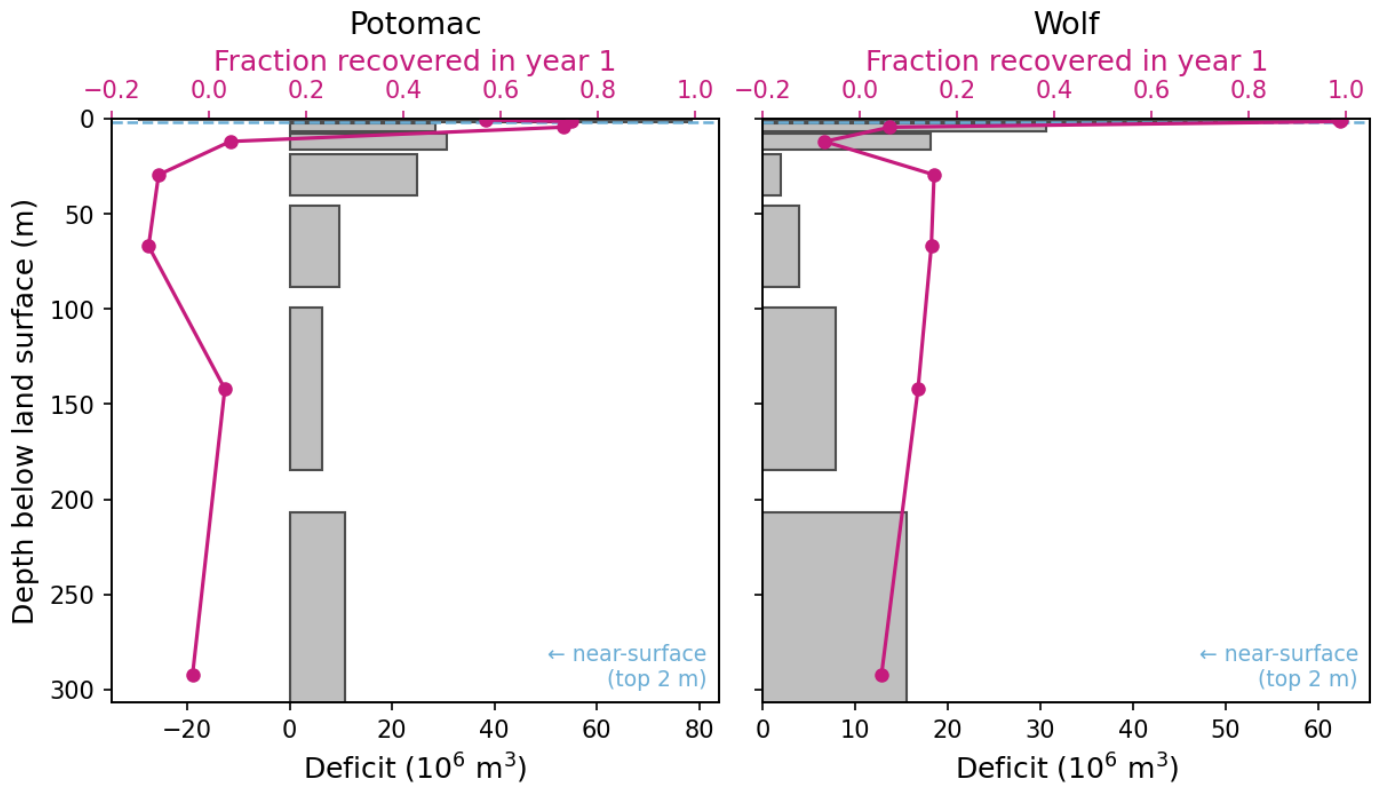
Script: analysis/shallow_deep_shielding.py

Aggregate: temporary ~ near-surface, persistent ~ deep

Domain	Drought	% of temporary in near-surface	% of persistent in deep
Potomac	1 / 3 / 10 / 50 yr	80 / 80 / 80 / 78%	43 / 52 / 74 / 92%
Wolf	1 / 3 / 10 / 50 yr	100 / 99 / 97 / 94%	94 / 97 / 98 / 99%

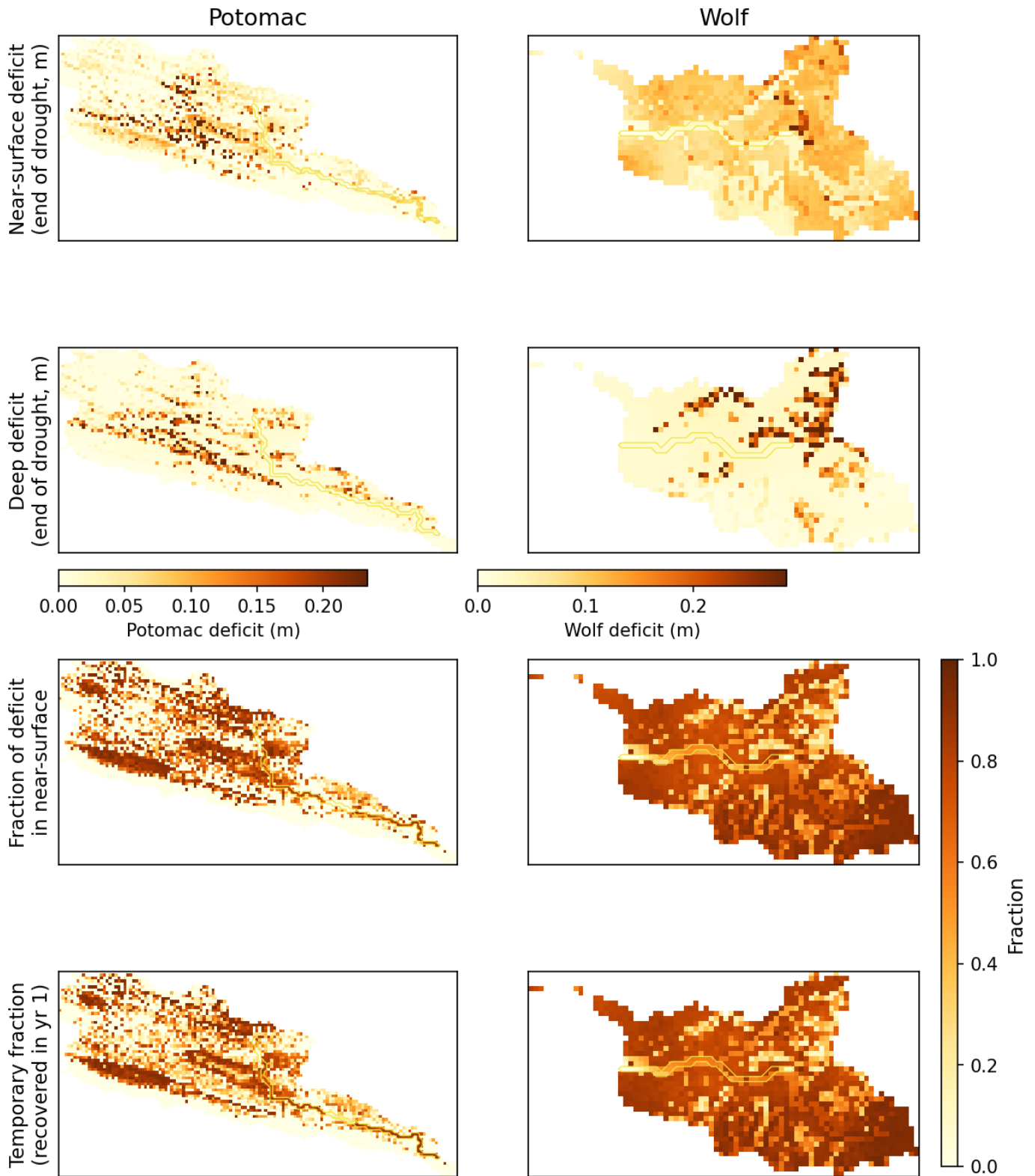
Temporary volume saturates by ~3-10 yr (Wolf ~125x10⁶ m³; Potomac ~80x10⁶ m³), almost all near-surface. Persistent volume keeps growing, and that growth is almost entirely deep (Wolf 50-yr persistent ~216x10⁶ m³ deep vs 3x10⁶ near-surface).





Year-1 recovery fraction ~ 1 in the top 2 m, $\ll 1$ below.

Spatial (10-yr)



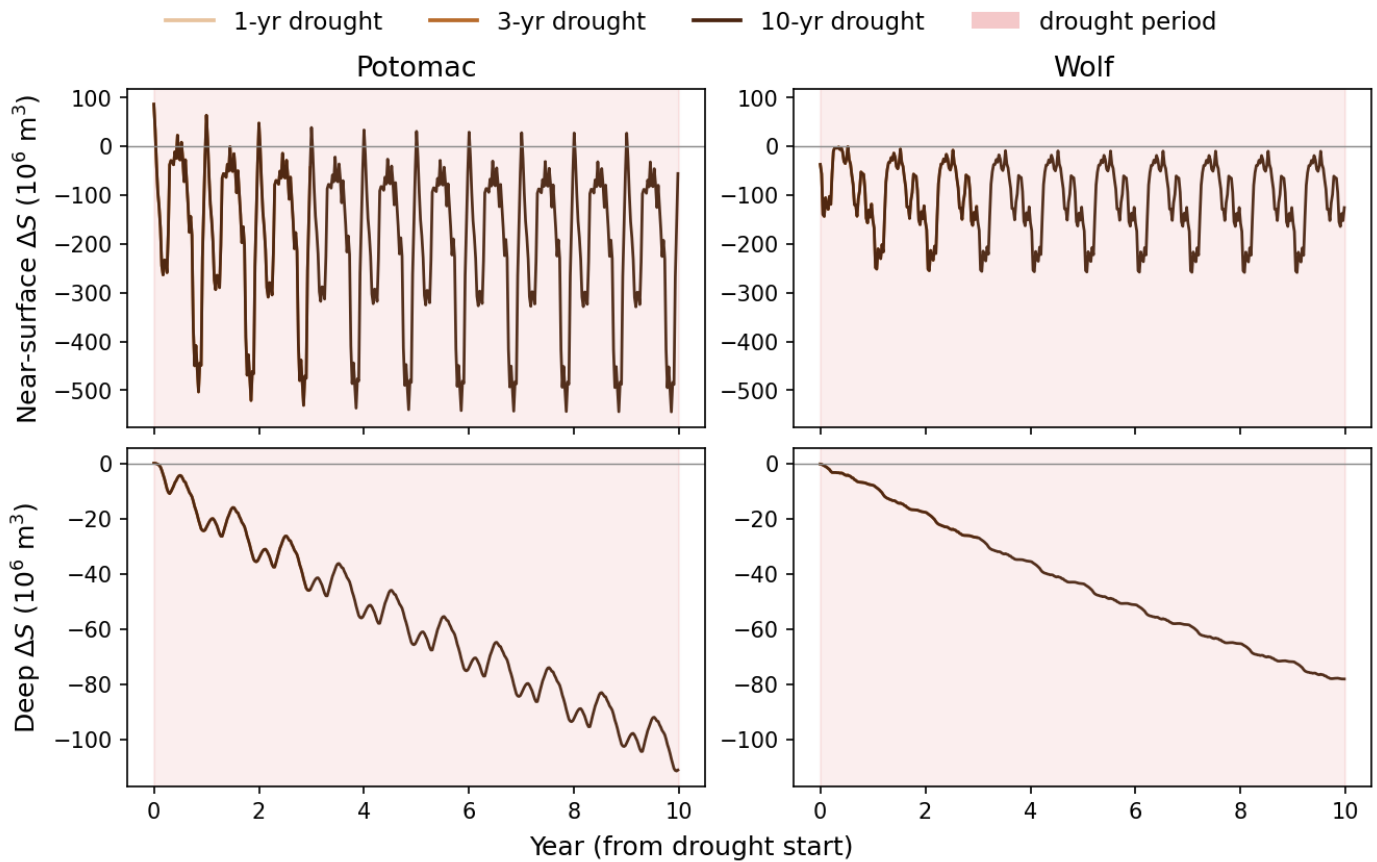
Bottom two rows are nearly interchangeable: fraction of deficit in near-surface vs temporary fraction. Spearman (non-stream cells with deficit):

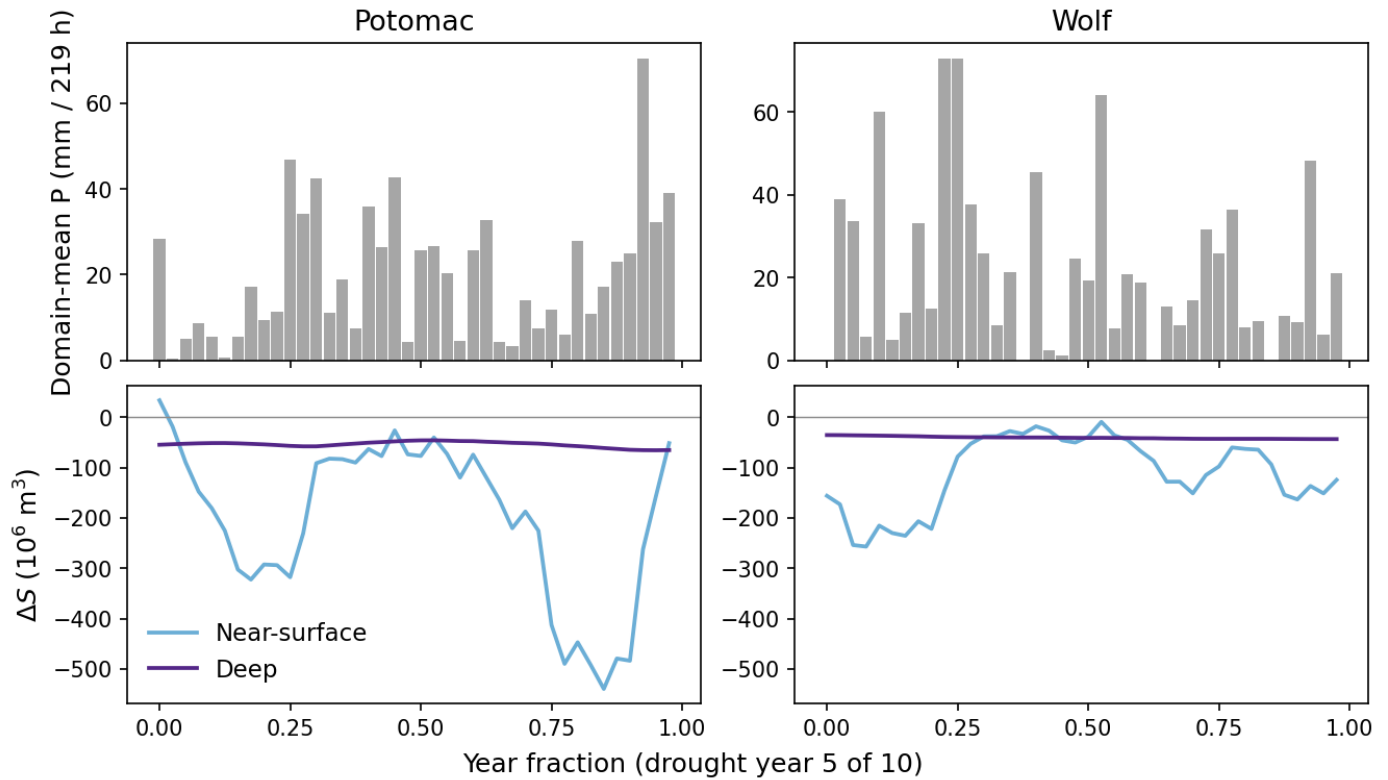
Domain	$\rho(f_{\text{near-surface}}, f_{\text{temp}})$	$\rho(\text{deep, persistent})$	$\rho(\text{near-surface, temporary})$
Potomac	0.85	0.93	0.84

Wolf	0.99	0.99	0.97
------	------	------	------

Wolf's map is mostly near-surface / temporary; Potomac's uplands are deep / persistent, with near-surface temporary limbs along flow-connected corridors - same geography as the Budyko local framing.

Mid-drought regeneration





Near-surface dS oscillates toward zero every year of the drought; deep dS ratchets down. In drought year 5 of 10, near-surface tracks P while deep stays flat.

10-yr pooled 219 h windows (stats; see also year-pulse figure):

Domain	Wet-window near-surface	dS	Wet-window deep	dS	$\rho(P, dS_{ns})$	$\rho(P, dS_{deep})$
Potomac	+72x106 m3	−0.2x106 m3			0.75	0.00
Wolf	+21x106 m3	−0.2x106 m3			0.68	−0.02

Read with Budyko framing

Local energy-limited / high-overland cells are the near-surface temporary limb: they "pay" drought with a regenerating skin of storage. Local water-limited / deep-WTD cells drain below 2 m; those losses never see the pulse recovery and become the persistent aggregate.